

Predicting Patient Self-reported Race From Skin Histological Images with Deep Learning

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INTRODUCTION

- Artificial Intelligence (AI) has advanced disease detection and biomarker prediction in computational pathology, but it also risks **learning hidden demographic biases** as a **shortcut**, achieving better performance but reduced generalizability.
- In pathology, race-related signals may arise from (i) **dataset biases** (e.g., batch effects or disease distribution) or (ii) true **biological differences** across races (e.g., tissue morphology).
- Skin histology provides a unique opportunity: while **melanin** and **pigmentation** are visually distinct in clinical dermatology, they are not obvious in H&E slides.

Goal

Investigate whether deep learning (DL) is able to predict **self-reported race** from **dermatopathology** slides and identify key **morphological correlates**.



METHODS

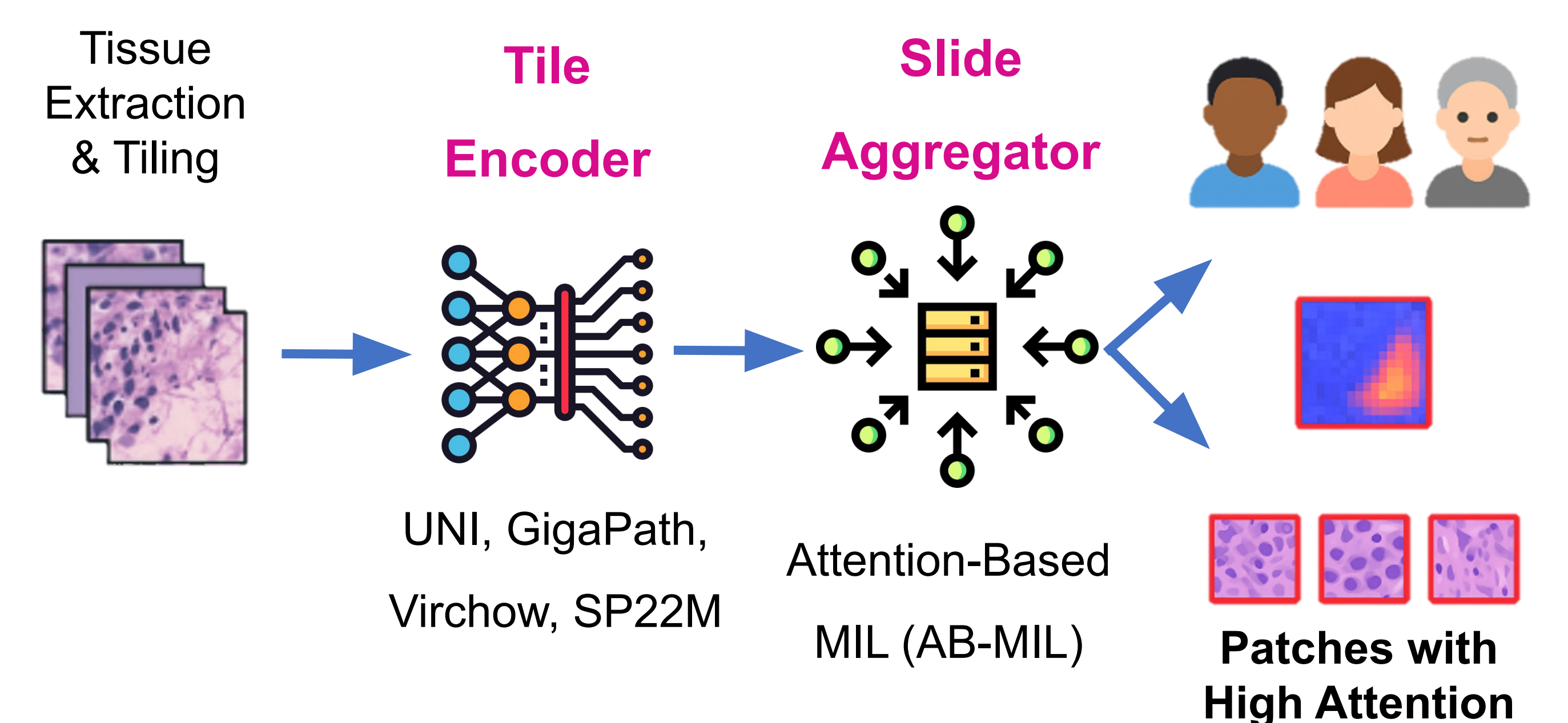
Dataset

All available skin specimens assembled from multiple sites in New York City (NYC) within Mount Sinai Health System (MSHS).

Self-reported Race	# Slides (%)	Patients %	MSHS %	NYC %	TCGA %
White	2,151 (40.8%)	39.3	43.1	31.2	73.7
Black	1,015 (19.3%)	19.0	21.7	29.9	10.3
Hispanic/Latino	868 (16.5%)	16.8	18.5	21.0	8.5
Asian	687 (13.1%)	15.7	10.3	5.7	7.1
Other*	543 (10.3%)	9.3	6.4	4.5	1.8
Total #	5,266	2,471	114,947	8M	23,276

Table 1. Summary of the skin dataset by self-reported race. *Other* includes American Indian or Alaska Native, Native Hawaiian or Other Pacific Islander. 'Unknown' is removed from analysis.

Model Learning & Interpretation



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RESULTS

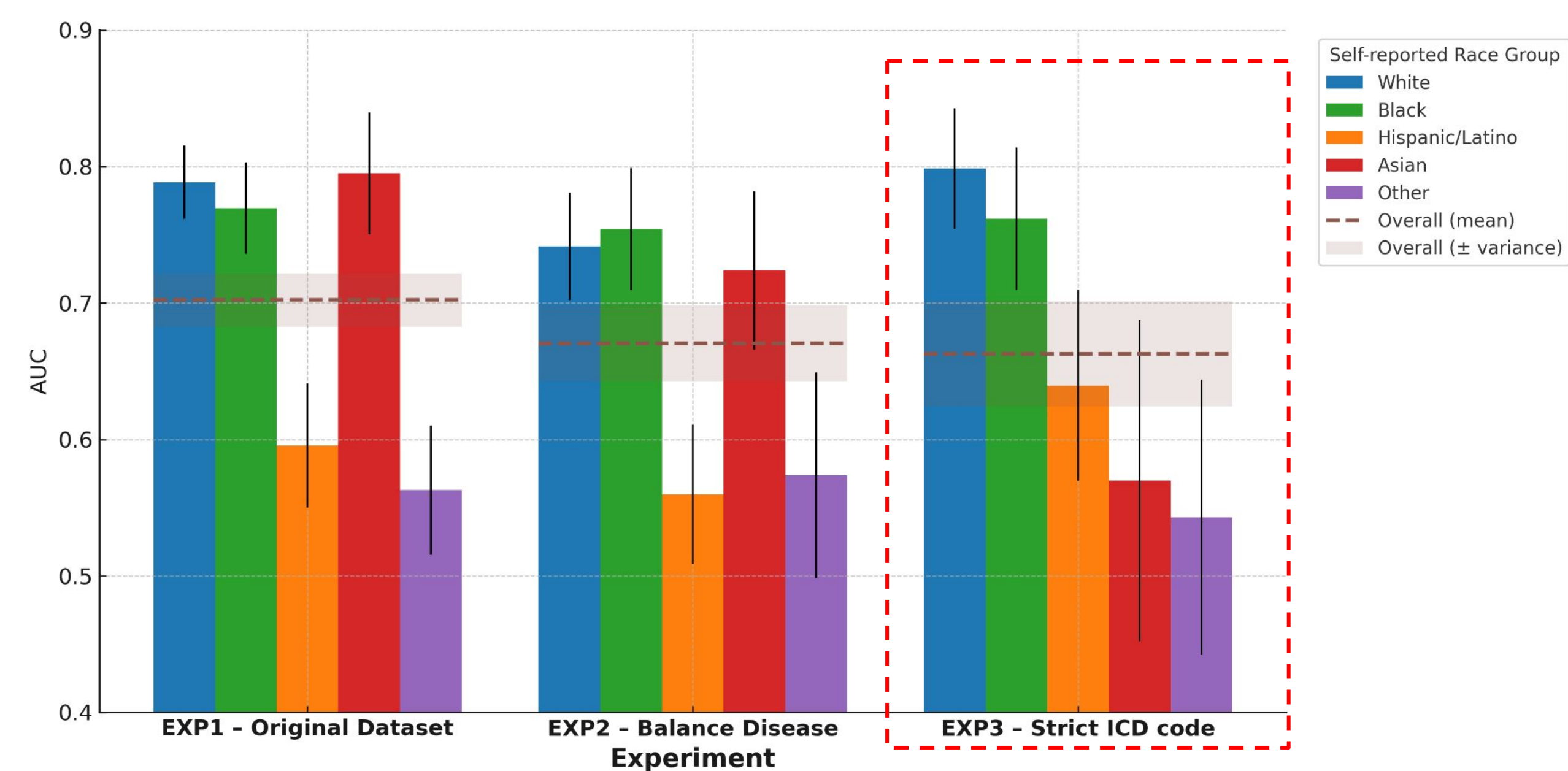


Figure 1. Model performance across three curated dataset. AUC is one-vs-all, accuracy is balanced accuracy

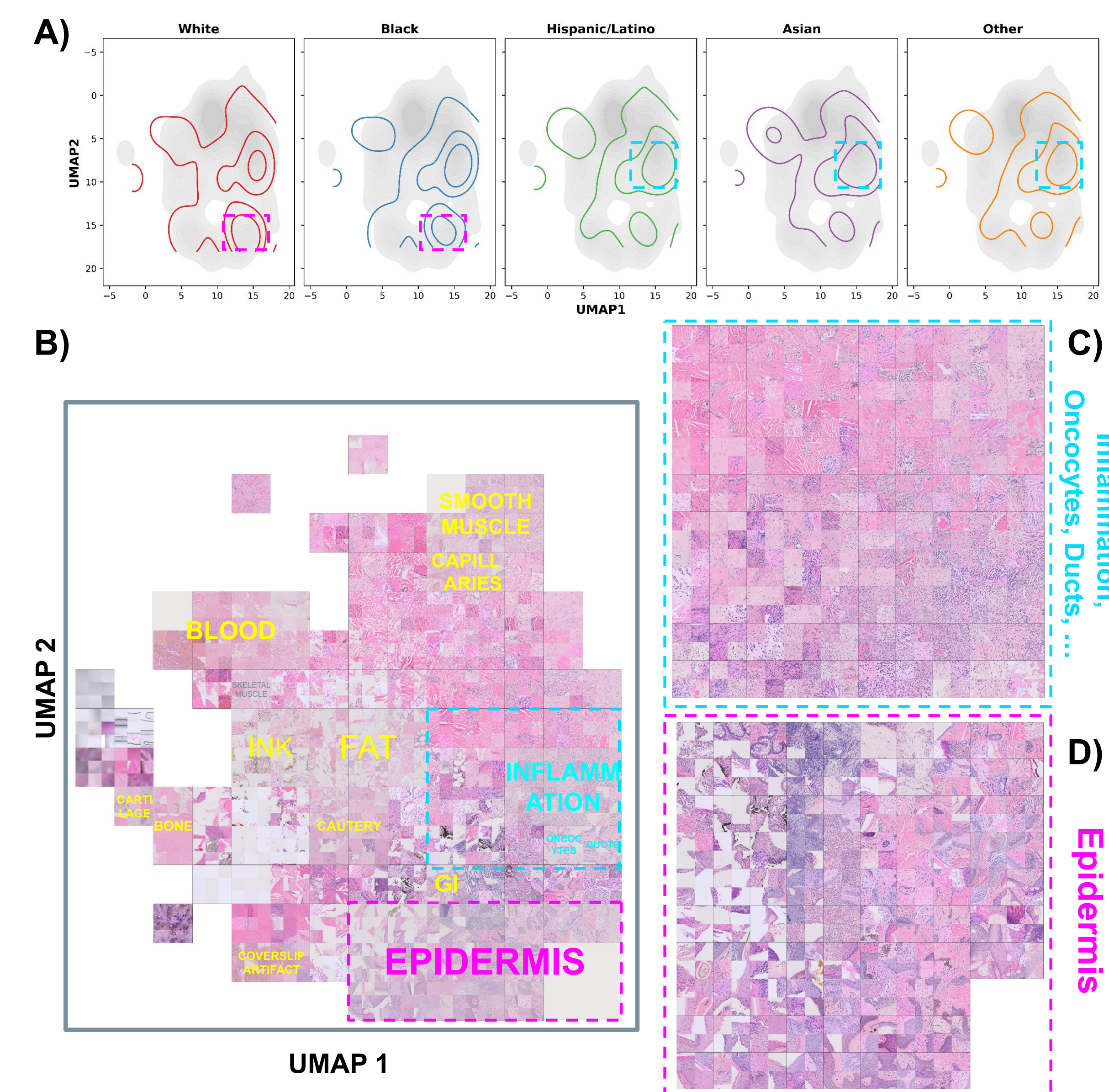


Figure 2. UMAP visualization of attention scores.

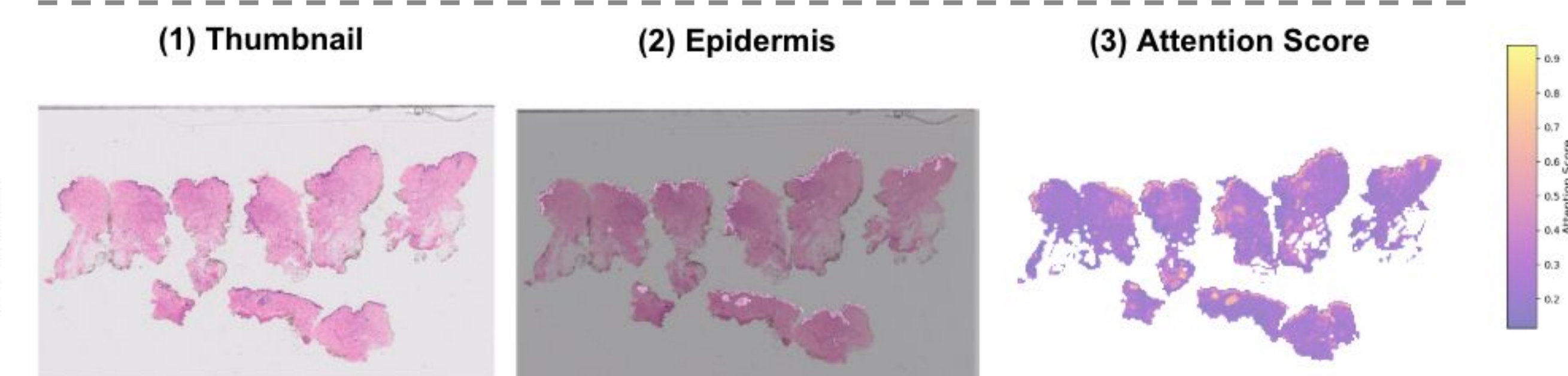


Figure 3. Whole-slide attention maps for selected example (White) from Exp3. Attention maps well to epidermis.

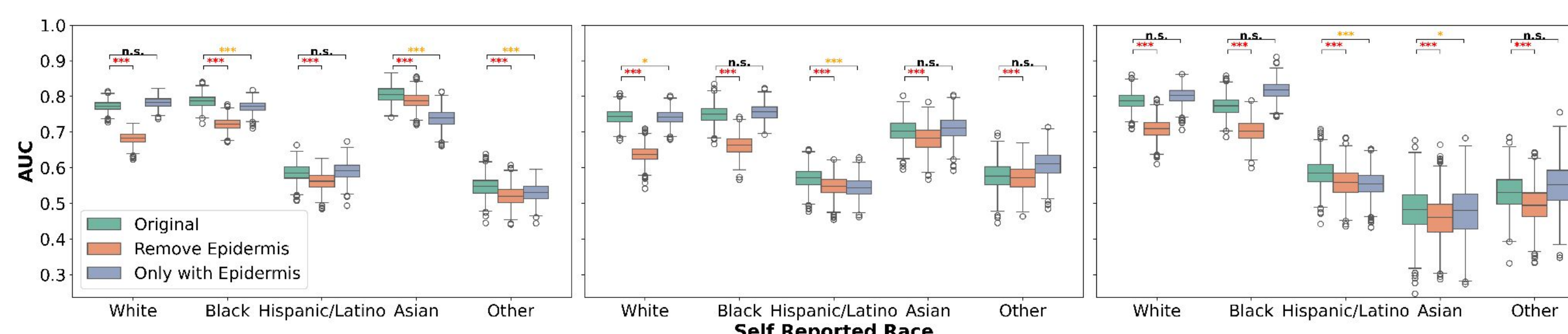


Figure 4. AUCs with epidermis tiles removed (orange), kept only (blue), and compared to original model (green)

CONCLUSIONS

- DL models can moderately predict self-reported race from skin histology, with **epidermal features** highlighted as potential correlates.
- Dataset composition affect model performance; imbalanced disease distribution across racial subgroups may inflate prediction performance.

